ELISA BUCCI

Mathematics Ph.D. Student | Quantitative Research & Machine Learning

Los Angeles, CA | ebucci@usc.edu

PROFILE

Mathematics Ph.D. student at USC researching statistical learning and random forests with Prof. Jinchi Lv. Strong foundation in optimization, probability, stochastic processes, and mathematical modeling, supported by coursework in mathematical finance and quantitative machine learning.

EDUCATION

University of Southern California (USC) | Los Angeles, CA — Aug 2024 - Present
Ph.D. in Mathematics

Rome Business School | Online — Apr 2025 - Apr 2026
International Master in Finance

Università di Pisa | Pisa, Italy — Oct 2021 - Jun 2024
M.Sc. in Mathematics | 110/110 cum laude

Università di Pisa | Pisa, Italy — Oct 2017 - Dec 2021
B.Sc. in Mathematics | 106/110

Selected coursework: Stochastic Calculus for Finance, Stochastic Differential Equations, Machine Learning for Quantitative Finance, Time Series, Game Theory (Economics & Finance), Optimal Control, Calculus of Variations, Numerical Analysis, Programming.

International study: Sorbonne Université (2023-24); Université Paris Cité (2022-23).

RESEARCH & SELECTED PROJECTS

Statistical Learning Theory: CART Random Forests — Ongoing

Ph.D. research at USC | Advisor: Prof. Jinchi Lv

- Exploring CART random forests from a stochastic-control perspective, interpreting feature subsampling as random action sets and tree splits as sequential allocation decisions.
- Investigating how split-allocation dynamics affect tree geometry and ensemble prediction risk. Research informed by Mei et al. (2026), arXiv:2605.26675.

Two Problems in Shape Optimization — 2024

Master's thesis | Supervisors: Profs. Emmanuel Trélat and Giuseppe Buttazzo

- Analyzed constrained optimization governed by elliptic partial differential equations using calculus of variations, optimal control, and relaxation methods.
- Studied optimality conditions, existence and uniqueness of minimizers, convergence, and shape/topological derivatives.

Hamiltonian Perturbation Theory and Mercury's Perihelion Precession — 2021

Bachelor's thesis | Università di Pisa

- Applied canonical transformations and action-angle coordinates to a restricted Sun-Jupiter-Mercury model, deriving a first-order analytical perihelion precession estimate of 154.65 arcseconds per century.

TEACHING EXPERIENCE

Teaching Assistant, Foundations of Statistics | USC — Aug 2026 - Present
Teaching Assistant, Calculus I-III | USC — Jan 2025 - May 2026
Undergraduate Mathematics Tutor | Università di Pisa — Mar 2023 - Jul 2024
Tutor, Mathematical Analysis I (Engineering) | Università di Pisa — May - Jul 2022
Private Mathematics & Physics Tutor | Independent — 2019 - Present

TECHNICAL SKILLS

Programming: Python, R, C, MATLAB | **Tools:** LaTeX, Excel

Methods: Statistical Learning, Random Forests, Constrained Optimization, Stochastic Control, PDEs, Numerical Analysis, Time Series

Languages: Italian (native), English (highly proficient), French (proficient)